Blockchain-Driven Automation for Degree Certificate Authentication while Preserving Data Integrity and ACID Properties

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***Abstract*—Academic records are constantly at risk of varying attacks which impede traditional methods because they implement manual processes and web services, something that SAC has already underlined as a threat. In this research, an endeavor has been made to investigate the way blockchain technology is benefiting certificate authentication and verification of degree certificates. This research paper develops a solution for using smart contracts on the Ethereum blockchain to prove the security, transferability, and validity of advanced certificates. This helps to validate student academic records securely. The information about the same is kept in a hashed pattern on a blockchain for the verification process so that it is tamper-free. Application of this method delivers a safe, reliable and scalable answer for schools to authenticate & handle academic credentials.**

Keywords—Blockchain, Ethereum, Smart Contracts, Hashing, ACID properties, keccak256, cryptography, decentralization

I. Introduction

Degree certificates are essential documents that confirm an individual's successful completion of a specific course of study. Traditionally, universities have relied on manual processes for issuing and authenticating these certificates, often managed by examination departments or student sections. However, the growing scale of higher education institutions and the increasing demand for swift and reliable verification procedures have highlighted the limitations of traditional methods. These limitations include inefficiency, susceptibility to errors, and potential vulnerabilities to fraud.

The growing dependence on digital technologies has revolutionized the way academic records are managed. This move towards automation and digitization has introduced new challenges concerning data security, integrity, and adherence to ACID properties. Upholding these properties is essential for guaranteeing the authenticity and reliability of academic records. Any breach can result in unauthorized access, data manipulation, or the generation of fraudulent certificates, which can ultimately undermine trust in academic credentials.

Blockchain provides a secure way to exchange any product, service, or business [1]. Although initially popularized by Bitcoin, it is the foundation of more than just cryptocurrencies [2]. The recent introduction of Blockchain has revolutionized the digital world, bringing new horizons for security, speed, and efficiency [3]. Blockchain technology is based on peer-to-peer collaboration, meaning it operates as a peer-to-peer network [4]. It has three main components: cryptographic keys, peer-to-peer networks, and digital ledgers. Each participant in the network has a unique set of cryptographic keys: a private key and a public key. The secret private key is used to sign the transaction, while the public key is used to verify the authenticity of the signature [5]. This digital signature acts as a unique and secure digital ID for you, ensuring that only authorized users can make or approve changes. Instead of relying on human oversight and control, it leverages networked computation to ensure trustworthiness [6]. This decentralized approach, often described as an "operating system for interactions" [7], empowers a system where trust is inherent in the technology itself, rather than dependent on individuals or institutions.

This research paper looks into using blockchain technology to make degree certificate authentication digital and automatic. It shows how blockchain can keep the ACID properties of the dataset safe. This implies that making degree certificate authentication automatic won't put the safety, honesty, or trustworthiness of school records at risk. Furthermore, this work displays how blockchain can cause a change in how schools handle degree certificates. It offers a safe, effective, and easy-to-grow answer for the future.

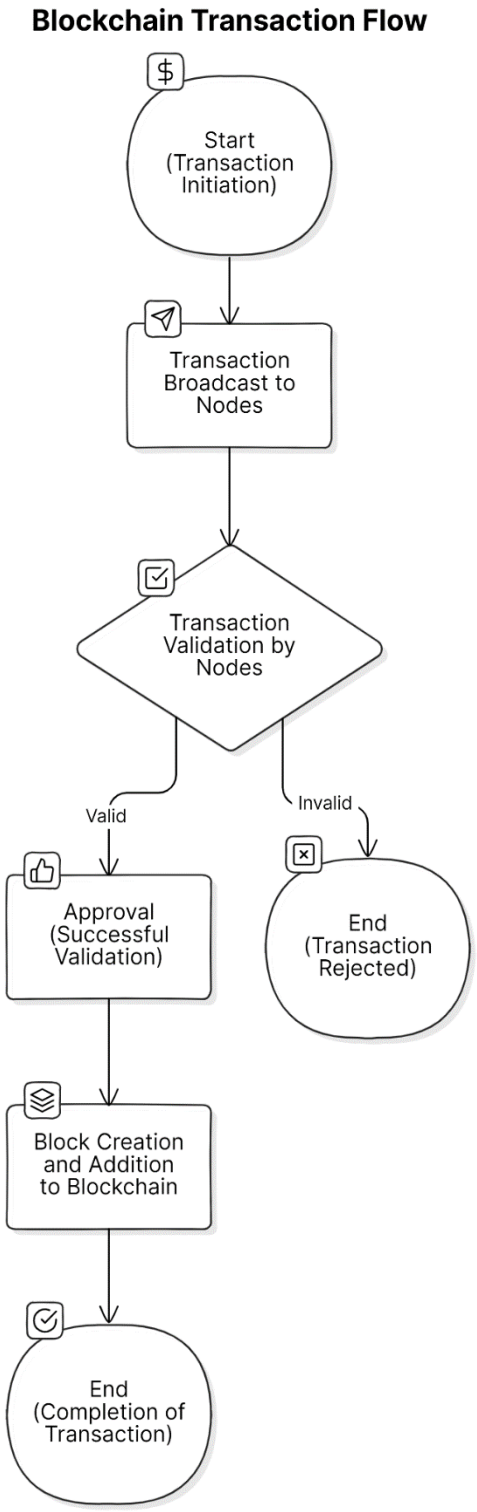

Fig. 1. Blockchain used for verifying and recording transactions [5].

Fig. 1 illustrates how blockchain technology is utilized to securely verify and record transactions. Each transaction is hashed and added to a decentralized ledger, ensuring immutability and transparency throughout the verification process.

II. LITERATURE REVIEW

This literature survey explores several innovative approaches that leverage blockchain to tackle these challenges, including the use of decentralized systems for revocation management, enhanced certificate transparency, and secure signature mechanisms This study focuses on current developments by examining different applications, evaluates their results, and identifies potential areas for further research in the field of blockchain-based authentication management [8].

A. Specific Application Based Review

1) Review on existing blockchain applications for certification

TABLE I. REVIEW ON EXISTING BLOCKCHAIN APPLICATIONS FOR CERTIFICATION

Paper Title	Work Done	Technique Used	Results
Blockchain and Smart Contract for Digital Certificate [9].	Proposed a digital certificate system to prevent forgery of academic certificates	Utilized blockchain technology to generate a key of the certificate data and stored it on the blockchain	Enhanced the credibility and security of certificates, enabling verification via QR code or website, reducing fraud and loss risks)
Certificate validation using blockchain [10]	Developed a system to store and validate digital academic certificates using blockchain.	Converted paper certificates to digital, used a chaotic algorithm to generate hash codes, and stored them on the blockchain	Provided a more secure and efficient method for certificate validation via a mobile application, simplifying the verification process for institutions and organizations
Explored the integration of blockchain technology into the educational sector [11].	Explored the integration of blockchain technology into the educational sector	Applied blockchain to manage and securely transfer student data, academic credentials, and educational databases	Provided a trusted verification system, allowing students to manage and share their achievements while enabling employers to access accurate, secure representations of students' potential
CertLedger: A new PKI model with Certificate Transparency based on	Proposed CertLedger, a new PKI architecture for certificate transparency	Utilized blockchain to prevent split-world attacks, manage certificates, and ensure	Improved TLS handshake performance and certificate validation,

Paper Title	Work Done	Technique Used	Results
blockchain [12]	using blockchain	certificate/revocation transparency	eliminating the need for traditional CA validation.
Blockchain Based Certificate Verification System Management [13].	Proposed a decentralized certificate revocation management and status verification system using blockchain	Extended the certificate structure to include a distribution point for revocation and used blockchain for management	Demonstrated improved performance in execution time and data consumption compared to traditional methods like CRL and OCSP, providing a more efficient and resilient solution
Certificate Transparency Using Blockchain [14].	Proposed a system called CTB to enhance CA accountability and certificate revocation using blockchain	Utilized blockchain (IBM Hyperledger Fabric) to ensure CA consent from domain owners for issuing certificates and included a revocation mechanism	Implemented a novel system with smart contracts in Go, improving certificate issuance transparency and revocation processes compared to traditional methods
ECBC: A High Performance Educational Certificate Blockchain with Efficient Query [15].	Developed the Educational Certificate Blockchain (ECBC) to enhance certificate management with better performance	Implemented a cooperative consensus mechanism, introduced MPT-Chain for efficient querying, and designed a privacy-protecting transaction format	Achieved low latency and high throughput, with efficient and quick query capabilities, outperforming traditional blockchain solutions in certificate management
Educert-chain: a secure and notarized educational certificate authentication and verification system using permissioned blockchain [16].	Proposed EduCert-Chain, a secure educational certificate authentication and verification framework using Hyperledger Fabric	Utilized ECDSA for digital signatures, SHA-256 for cryptographic hashing, and Raft consensus mechanism	Achieved an average throughput of 34.95 TPS (query) and 32.23 TPS (open), with average latencies of 4.21 s (query) and 2.05 s (open)

Table I. provides an overview of various blockchain applications used for certification management, highlighting key features, advantages, and limitations in existing systems. It also outlines areas for improvement, forming the basis for the research gaps addressed by our proposed solution

B. *Research Gap*

1. Necessity of common, accepted standards & protocols to enable the smooth functioning of blockchain certificate systems across various institutions and countries, enabling wide adoption as well as integration.
2. While the current iteration is an improvement, blockchain still requires more work to accommodate increased transactional load and decreased lag time overall.
3. Less examination of how blockchain technology can be mixed with and layered on top of existing educational databases and legacy systems.
4. Comprehensive solutions are thus needed to adequately safeguard user data and ensure a rigorous certificate management system remains transparent, secure over the long term.
5. High operational costs and resource consumptions associated with the existing implementations, further research should investigate ways to minimize costs and consumptions.

Thus, this literature survey highlights critical gaps in blockchain-based certificate management. Our project addresses these gaps by implementing a solution that utilizes Ethereum blockchain technology to manage certificate issuance, verification, and updates. By employing keccak256 hashing for secure data handling and integrating with existing systems for efficient data processing, our approach ensures enhanced scalability and performance. Additionally, our system facilitates easy integration with existing systems, preserves user privacy through secure hash storage and minimizes costs through optimized blockchain interactions. This comprehensive approach not only advances blockchain applications in certificate management but also offers a practical solution to the challenges identified in current research.

III. METHODOLOGY

The methodology section outlines the technical approach and implementation of our blockchain-based educational certificate management system [17]. Our system aims to create a secure and tamper-proof system of issuing, storing (on the Ethereum blockchain) and authentication of education certificates using built-in smart contracts.The system utilizes the keccak256 (SHA-3 family) as the hashing algorithm to ensure the integrity of certificate data.It is designed to handle bulk uploads of certificate data from universities and single-entry uploads, allowing for flexibility in managing data. This allows educational facilities with established infrastructures to implement this technology without the need for significant changes. This system solves the common issues of certificate forging, centralized control and long verification process by combining the blockchain's decentralized & immutable nature with Smart Contract's flexibility.

A. *System architecture overview*

Our blockchain-based certificate management solution is specifically designed with an architecture to securely, efficiently, and verifiably store educational certificates. The

input provided starts with bulk uploading of certificate details via CSV or individual entry. Further, verifiable fields include PRN (Personal Registration Number) which is the equivalent of a unique ID provided to each student by their university, Name, Degree, Course Name, Year of Graduation, and CGPA. Once the data is provided, a hash is created via the keccak256 algorithm, which creates a single, unique cryptographic hash per certificate.

It is first stored in a secure database and then sent to the blockchain upon the execution of a smart contract [18]. The smart contract would then validate the data, add the certificate information to the blockchain, and then update, making sure that the information does not change. Stored, the hash can then be obtained and rehashed to check against the hash of the submitted certificate for verification.

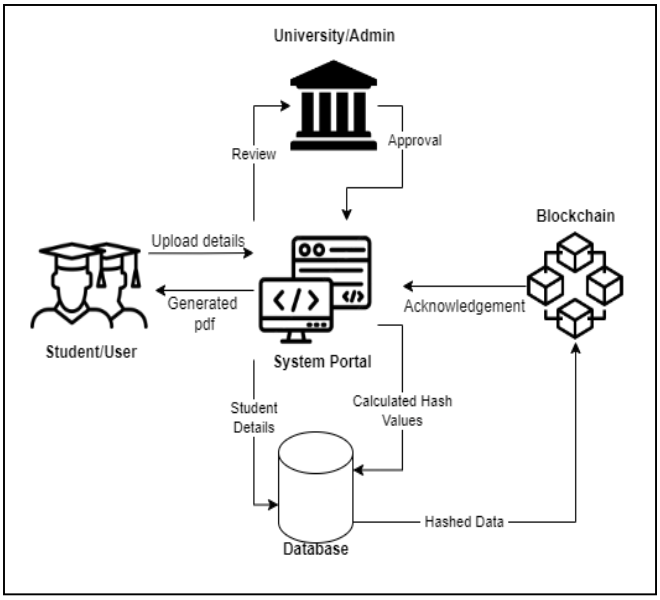

Fig. 2. System Architecture

Fig. 2. visually represents the architecture of the system, outlining the flow of data from input (certificate details) through to the hashing process, storage on the blockchain, and subsequent verification. The process includes:

1. User inputs certificate data.
2. Data is parsed and hashed using the Keccak256 algorithm.
3. The admin verifies the data from the university database and sends an approval if the data is valid.
4. The hashed information is first stored in the database for backup along with the quick retrieval and then written to the blockchain.
5. Data is verified by comparing stored and new hashes.
6. Generated certificate is displayed to the user.

This architecture ensures data integrity, security, and verifiability while maintaining a seamless process for certificate issuance and verification.

B. *Data Flow and Process Workflow*

1) Data Parsing and Hashing : Data flow in the system begins with the input of certificate details, which is capable of being in two ways: a bulk upload using a CSV file, or by single entry. In the case of bulk uploads, a CSV containing multiple entries is parsed and all the basic certificate information is extracted, like PRN, Name in both English and Devanagari, Degree, Course Name, Year of Graduation, and CGPA. In the case of a single entry, the user manually fills in the same details into the system.

Upon receiving the data, it gets automatically parsed and processed by the system for further operations. The next most crucial step is hashing the data. The Keccak256 algorithm of the SHA-3 family will then be used to generate a unique cryptographic hash corresponding to every certificate. This step of hashing guarantees that tampering with the info within the certificate is impossible seeing that any alteration to the enter statistics will completely alter the hash. This specific hash acts as a virtual fingerprint for every certificates.

2) Storage and Verification : After the certificate data is hashed, the system proceeds to store the hash securely. The storage occurs in two locations: first, the hash is stored in a centralized database for quick access and backup purposes. Then, the system transfers data to the blockchain via smart contract execution, which handles functions such as adding, updating, and validating certificates. As for the verification method, the system rehashes the certificate data supplied through the person and compares the newly hashed cost with the only recorded on the blockchain after a verification request is made by way of both a student or public verifier.. If both the hashes match, the certificate is valid and unaltered. It grants immutability to the records and presents a decentralized way of verification through blockchain technology.

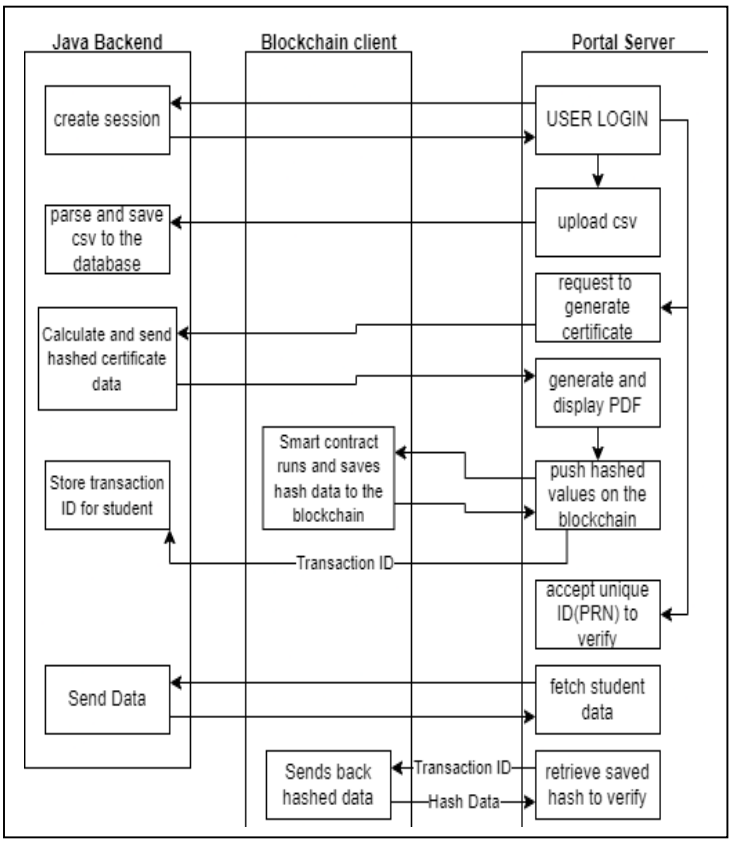

Fig. 3. Complete System Process Workflow

Fig. 3. represents the step-by-step workflow :

1. *User Login:* User (university admin) logs into the system.
2. *Uploading Data*: Certificate data is uploaded (via bulk CSV or single entry).
3. *Parsing Data:* The system parses the certificate details.
4. *Hash Generation:* Keccak256 algorithm generates the hash for each certificate.
5. *Storing Hash:* Hash is stored in the database and transferred to the blockchain via smart contracts.
6. *Verification:* A rehash of certificate details is generated and compared with the stored hash for verification.
7. *Certificate Printing:* Upon successful verification, the certificate is available for printing and verification.

C. *Smart Contracts and Blockchain Integration*

Smart Contracts are executable codes that run on top of the Blockchain [19]. As opposed to traditional contracts, smart contracts allow users to encode their agreements and trust relationships. Its primary function is permitting automated transactions with out the involvement of a government.. To safeguard against tampering, copies of smart contracts are distributed across all nodes in the blockchain network. By automating these operations, blockchain platforms help reduce human error, minimizing disputes related to contract execution. Smart Contracts automate processes within a network and have transformed traditional paper contracts into digital ones. [20]. In the context of the Ethereum blockchain, these contracts are written in Solidity, a specialized programming language designed for such contracts [21].

For this project, smart contracts on the Ethereum blockchain are implemented using Solidity [22]. Fundamentally, Solidity is intentionally made to build smart contracts by which complex logic can be encoded into simple and executable instructions. Here, three core functionalities are being governed by these smart contracts: certificate validation, addition, and updates.. These functions handle the creation, storage, and verification of the certificates securely on the blockchain. These operations are written within a single Solidity-based smart contract that runs autonomously once deployed on the Ethereum network.

The workflow for the proposed blockchain-based certificate management system can be broken down into multiple stages, each playing a crucial role in ensuring the integrity, security, and transparency in the process of generating and validating certificates. Starting with the submission of certificate data, the system uses cryptographic hashing through the Keccak256 algorithm to create unique, tamper-proof digital signatures. These hashed signatures are then stored on the Ethereum blockchain via smart contracts, making the certificates publicly verifiable. Fig.4 visually represents this end-to-end process.

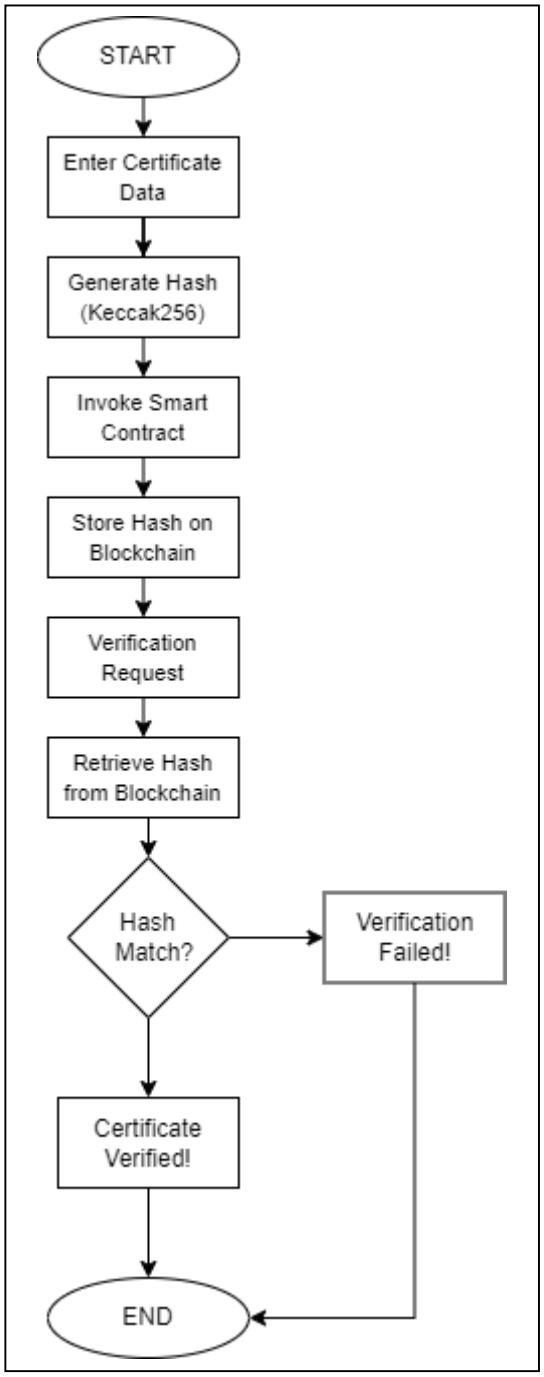

Fig. 4. Flowchart of Certificate Hashing and Smart Contract Interaction

Fig. 4. depicts the step-by-step process involved in generating a certificate hash and interacting with the smart contract on the blockchain. This flowchart outlines the key stages, including data input, hashing using the Keccak256 algorithm, and the subsequent storage of the hash in the blockchain for secure and verifiable certificate management.

D. Hashing using the Keccak256 algorithm

The Keccak[r,c] sponge function works using the parameters capacity (c) and bitrate (r), with the assumption that for simplicity, r is a multiple of the lane size, which is commonly the case for standard implementations [23]. The input message (M) is represented as a string of bytes (Mbytes) and may include a few trailing bits (Mbits) for domain separation. These trailing bits are encoded as the delimited suffix (d). The input is padded to form a message (P), which is organized into blocks (Pi) and further arranged as arrays of lanes. The state variable (S) holds these arrays, and the || operator is used for string concatenation throughout the process [23].

ALGORITHM I KECCAK[R,C] SPONGE FUNCTION PSEUDO-CODE [23].

```
Keccak[r,c](Mbytes || Mbits) {
  # Padding
  d = 2^|Mbits| + sum for
i=0..|Mbits|-1 of 2^i*Mbits[i]
  P = Mbytes || d || 0x00 || … || 0x00
  P = P xor (0x00 || … || 0x00 ||
0x80)

  # Initialization
  S[x,y] = 0,
for (x,y) in (0…4,0…4)

  # Absorbing phase
  for each block Pi in P
    S[x,y] = S[x,y] xor Pi[x+5*y],
for (x,y) such that x+5*y < r/w
    S = Keccak-f[r+c](S)

  # Squeezing phase
  Z = empty string
  while output is requested
    Z = Z || S[x,y],
for (x,y) such that x+5*y < r/w
    S = Keccak-f[r+c](S)

  return Z
```

Here, the Keccak256 algorithm securely hashes certificate data into a fixed-length string of 256 bits from key attributes such as PRN, student name, degree, course name, year of graduation, and CGPA. This hash is unique and serves as the digital fingerprint of each certificate. As the algorithm is immutable and unique in nature, even the slightest change in the certificate information will lead to a completely unique hash value. The process starts with collecting the certificate details, either through bulk upload (CSV) or single entry. These details are parsed, concatenated into a single string, and then fed into the Keccak256 algorithm. The resulting hash is a compact and irreversible representation of the original data which is subsequently stored on the blockchain via the smart contract.

TABLE II. SAMPLE CERTIFICATE DATA AND HASH VALUES

PRN	Name	Degree	Course	Year	Cgpa	Keccak 256 Hash
		B.E.	Computer Science	2023	8.9	0xA0F 9....
		B.A.	Economics	2022	7.5	0xB1A 7..
		M.Tech	Mechanical	2024	9.2	0xC9E 3...
		B.Sc.	Information Tech	2021	8.3	0xD2B 9...
		M.B.A.	Finance	2020	7.8	0xF5A. ..

Table II. presents a collection of student data entries along with their corresponding hash values generated using the Keccak256 algorithm. This table illustrates the relationship between the original certificate details and their unique digital signatures, emphasizing the secure and verifiable nature of the certificate generation process.

E. Verification

The system at the time of certificate issuance pushes the Keccak256-generated hash of the certificate details onto the blockchain. In the verification process, the same certificate details are fed by the verifier, e.g., PRN, name, degree, etc. Once again, the same set of details is passed through the Keccak256 algorithm to produce a new hash value. The system compares this generated hash with the one that was originally stored on the blockchain at the time of issuance.

The verification process in the proposed system hinges on comparing two cryptographic hashes generated using the Keccak256 algorithm. Let H(m) denote the original hash derived from the certificate data 'm' (e.g., PRN, name, degree, etc.) when the certificate was initially issued and stored on the blockchain. During verification, a new hash H'(m) is computed from the certificate details provided by the user. The system then compares the two hashes

$$H(m) == H'(m) \tag{1}$$

The comparison works as follows:

1. If the newly generated hash matches the stored hash, it confirms that the certificate data has not been altered since issuance, and the certificate is deemed authentic.
2. If the hashes do not match, the system flags the certificate as invalid, indicating possible tampering or incorrect data input.

Thus the methodology section discusses the steps involved in setting up a secure and verifiable system in educational certificate management through the use of blockchain technology. Starting from generating hashes based on Keccak256, data involved in each certificate is hashed securely to produce a unique and tamper-resistant digital signature. Smart contracts stored on the Ethereum blockchain automate storage and verification processes for certificates, meaning the data is immutable and transparent. It takes care of the whole flow of data related to any issuance and verification process, allowing for bulk and single entries without much hassle. The final verification process with its cryptographic hash comparison is also capable of spotting tampering or forgery attempts very easily. Overall, the methodology ensures a secure, scalable, and reliable framework for digital certificate issuance and verification, protecting both institutions and individuals from fraudulent practices.

IV. FUTURE SCOPE

The system can be enhanced in future by implementing additional features and improving existing ones. One potential enhancement is the introduction of a student login system, allowing students to directly access their profiles and monitor the progress of their degree certificates. Enhancing security through biometric authentication for the Controller of Examination (CEO) is another feature that we can incorporate to provide access control. Moreover, introducing multi-language support, including regional languages, can enhance the system's usability, especially for users from diverse backgrounds. Similarly, expanding email functionality to enable bulk communication with students can streamline interaction. In a similar fashion, facilitating dynamic notifications can improve overall engagement.

Future development can also focus on more advanced analytics and data visualization tools within the dashboard to offer deeper insights into student performance. Cloud-based deployment could provide scalability, flexibility, and ease of maintenance, making the system capable of handling larger datasets and user volumes efficiently. In addition, the integration of external verification systems would ensure the authenticity and security of certificates, preventing tampering and enabling real-time verification by third parties. Developing a mobile application would further improve accessibility, enabling students and administrators to interact with the system on the go. By prioritizing these aspects, the system can continue to evolve to meet both current and future operational demands.

V. CONCLUSION

Thus, in this paper, we have proposed an all-inclusive blockchain-based framework to securely manage educational certificates, aiming to address the pervasive issue of certificate forgery and fraud in this domain. We ensured the immutable and publicly verifiable data about certificates through the usage of the Ethereum blockchain and the Keccak256 hashing algorithm. Using smart contracts, the system is able to issue, store, and verify certificates with security at its core; hashed certificate data is also logged on the blockchain, providing long-term integrity and ease of decentralized verification through comparison of the new hashes against hashes stored on the blockchain. We also made the solution adaptable by including Java for having hitch-free integration with legacy systems and therefore has the ease of integration within the current infrastructure. Our approach emphasizes on the security of the system in a scalable and effective way through handling unapproved manipulation of certificates. All this is in a framework with the key combination of cryptographic hashing and decentralized storage along with automation to

give it a future-proof status towards overcoming the challenges presented by digital certificate management.

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